

§2.3

Other Displays of Data

The right picture for the data you have

Today's Goal

- Create and interpret dotplots and stem-and-leaf plots
- Distinguish between appropriate uses of bar graphs and histograms
- Identify and interpret pie charts and Pareto charts
- Use Excel to create line graphs, bar graphs, and pie charts
- Recognize deceptive graphical displays and their misleading features

What You Will Be Graded On

LT.5 Graphical Summaries of Data

I understand the different types of graphical displays for data (e.g., bar graphs, histograms, boxplots, etc.). I can interpret and create appropriate graphical summaries for various types of data.

Twenty Numbers, What Picture?

Twenty adults had their resting heart rate measured, in beats per minute.

72 64 86 66 58 70 63 75 66 81

61 78 64 55 69 72 66 74 71 68

What picture would you draw?

A histogram is one answer. Today you get nine more, and the real skill is choosing.

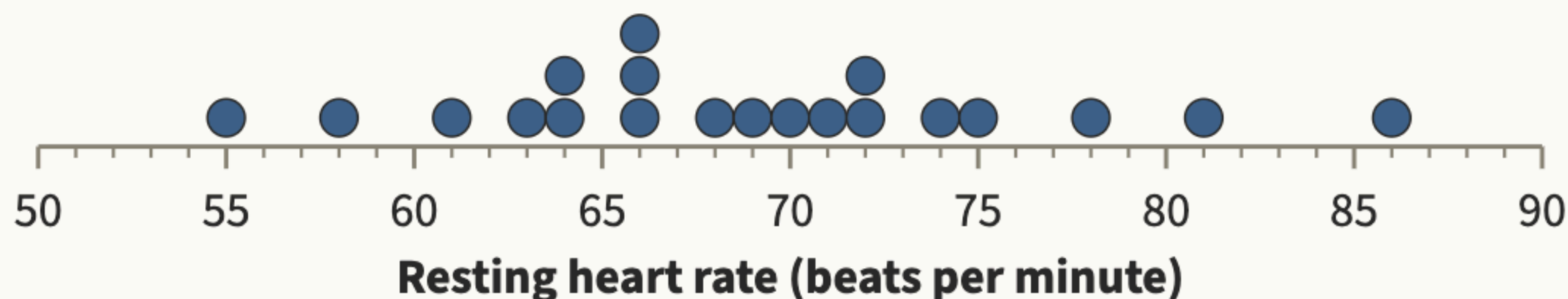
Every Value Gets a Dot

Definition 2.3.1

A **dotplot** is a graph of quantitative data in which each data value is plotted as a point (or dot) above a horizontal scale of values. Dots representing equal values are stacked.

Draw a number line. Put one dot per value. Stack the repeats. That is the whole recipe.

Reading a Dotplot



(a) Describe the shape.

Approximately symmetric, with one peak in the 60s.

(b) How many adults had a heart rate of 66?

3, the tallest stack.

(c) What is the most common range?

The 60s. Nine of the twenty dots sit there.

Split Each Value in Two

Definition 2.3.4

A **stem-and-leaf plot** (or **stemplot**) is a graph of quantitative data in which each data value is separated into two parts:

- the **leaf** is the last significant digit in the data value, and
- the **stem** is the rest of the data value.

Each stem appears on its own row with the associated leaves listed to the right of the stem.

Same Data, New Picture

The twenty heart rates from Example 1.

(a) What values are in the first row?
55 and 58.

(b) Smallest and largest heart rate?
55 and 86: first leaf, last leaf.

(c) Turn your head sideways. What do you see?
The dotplot's shape, with every value kept and sorted.

Stem	Leaves									
5	5	8								
6	1	3	4	4	6	6	6	8	9	
7	0	1	2	2	4	5	8			
8	1	6								

Connect the Dots

Definition 2.3.7

A **line graph** is a visual representation of data that consists of points connected by straight line segments. The horizontal axis typically represents the independent variable such as time or categories, while the vertical axis represents the dependent variable such as quantities or measurements.

Time Along the Bottom

Definition 2.3.11

A **time-series graph** is a graph of time-series data, which are quantitative data that have been collected at different points in time.

It is a line graph with time on the horizontal axis. It is the graph you see in the news every day.

Trends Over Time

A campus café counted its iced coffee sales each month for a year.

(a) Busiest month, and about how many?

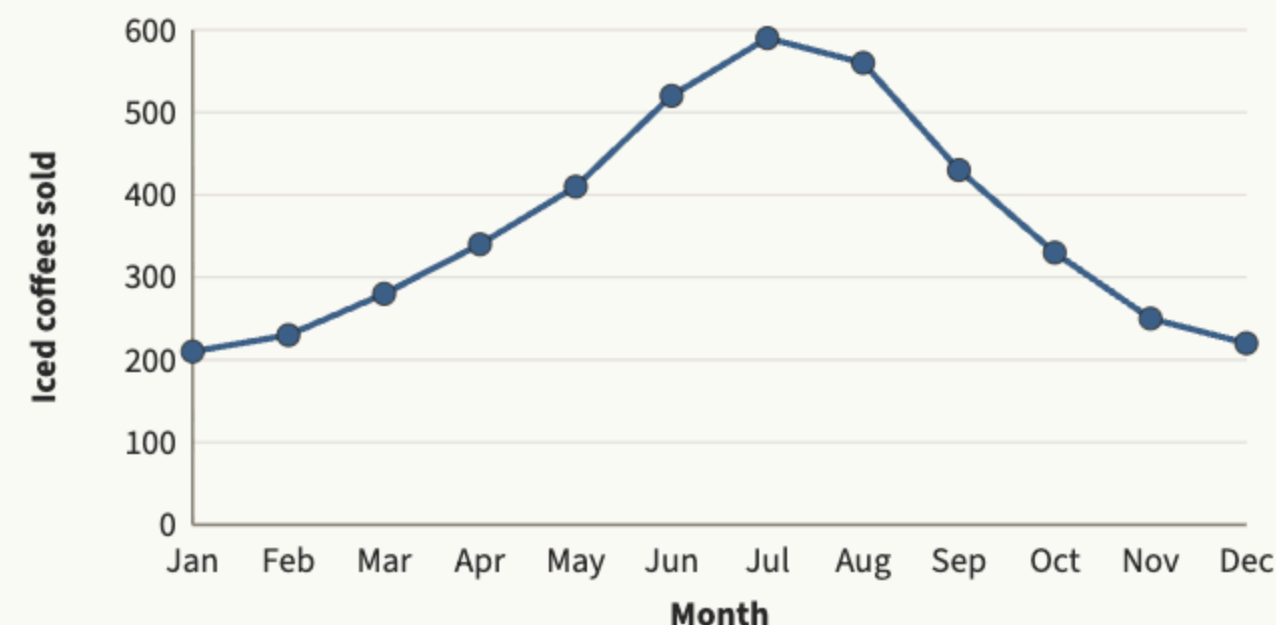
July, about 590.

(b) The trend from January to July?

Up every month, slowly and then fast.

(c) About how many in October?

About 330, read off the gridlines.



Bars for Categories

Definition 2.3.17

A **bar graph** is a graph of bars of equal width to show frequencies (or relative frequencies) of categories for categorical (or qualitative) data. The bars may or may not be separated by small gaps.

Categorical data. That one word is the whole difference from a histogram.

Bar Graph or Histogram?

Both are bars. Both show frequencies. Students mix them up all the time.

What decides which one you draw?

	Bar graph	Histogram
Data	Categorical (qualitative)	Quantitative
Bars	May have gaps	Touch, unless the data has a gap
Order	Any order you like	Follows the number line

How Did You Get to Campus Today?

Sixty students in a statistics class answered.
Bars are in alphabetical order.

(a) Which way was most common?

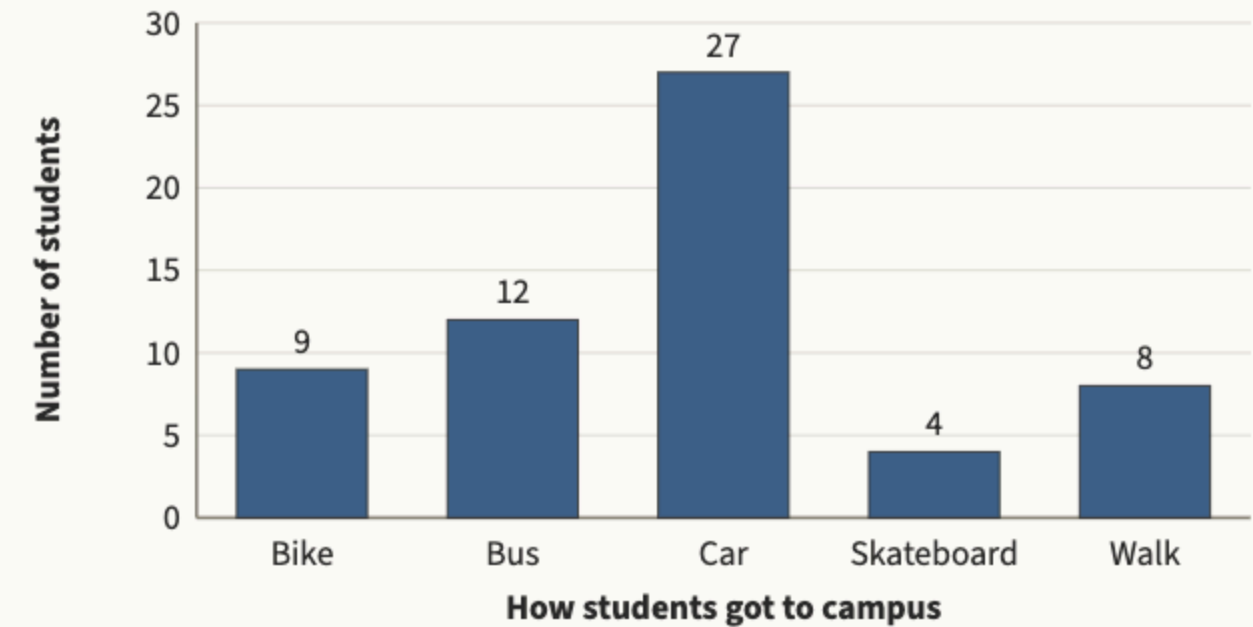
By car: 27 of the 60.

(b) How many came by bus?

12.

(c) Why not a histogram?

“Bike” and “Car” are categories, not numbers.



Tallest First

Definition 2.3.21

A **Pareto chart** is a bar graph in which the categories are arranged in descending order of frequency so the bars decrease in height from left to right.

Same bars as a bar graph. Only the order changes, and the biggest category jumps out.

The Same Sixty Students, Reordered

Example 4 again, tallest bar first.

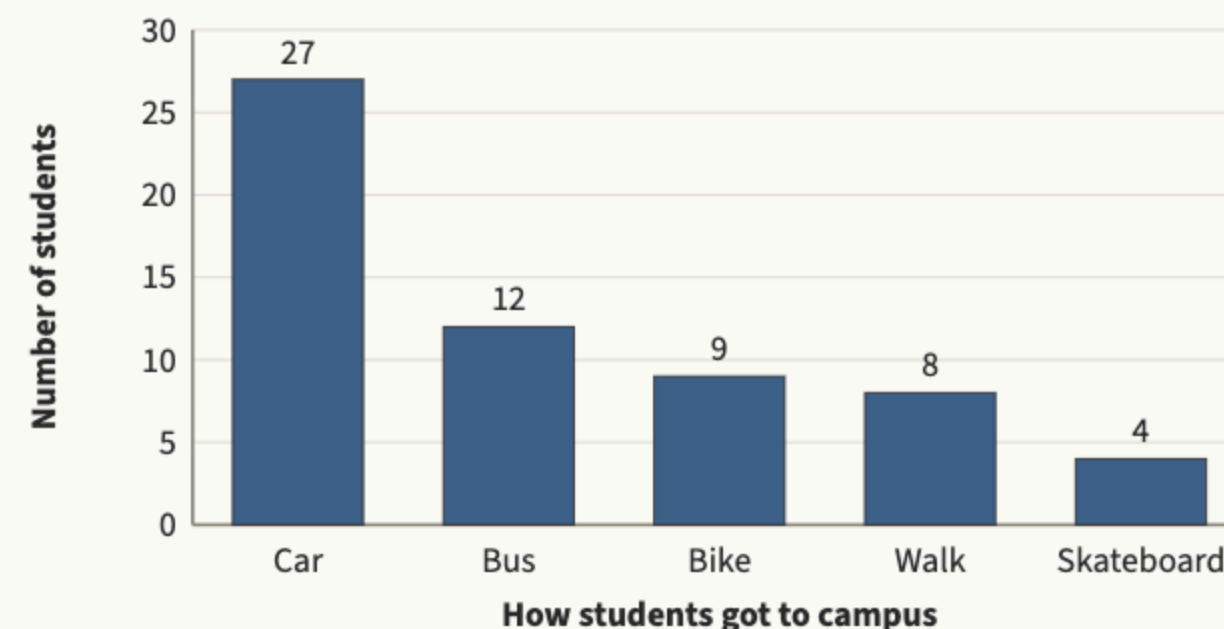
(a) The campus wants fewer car trips.

Where to start?

Car. Nothing else comes close.

(b) Which two categories make up the majority?

Car and Bus: $27 + 12 = 39$ of 60, or 65%.



Slices of One Whole

Definition 2.3.24

A **pie chart** is a graph that depicts categorical data as slices of a circle, in which the size of each slice is proportional to the frequency count for the category.

Good for parts of one whole. Weak when two slices are close in size.

The Same Sixty Students, as a Pie

Example 4 a third time. Percents are rounded to whole numbers.

(a) Which category is the largest slice?

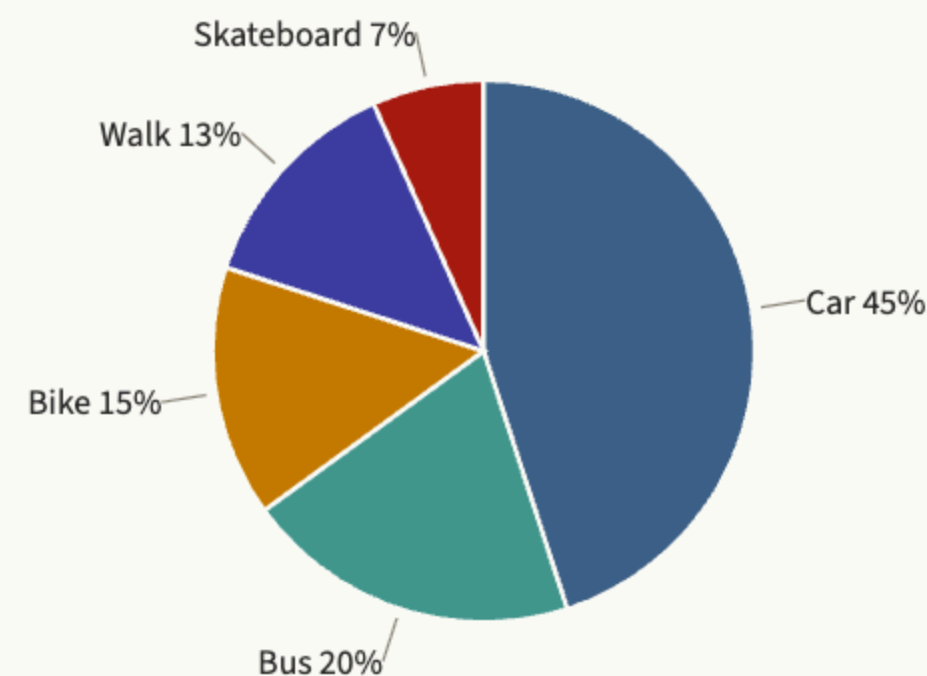
Car, just under half the circle.

(b) About what percent came by car?

45%, which is 27 out of 60.

(c) Why is Bike versus Walk hard to compare?

Nearly equal slices. The bar graph shows 9 versus 8.



Now You Try: Read the Pie

You Try 2.3.2. 250 students said where they ate lunch yesterday.

(a) Most common answer?

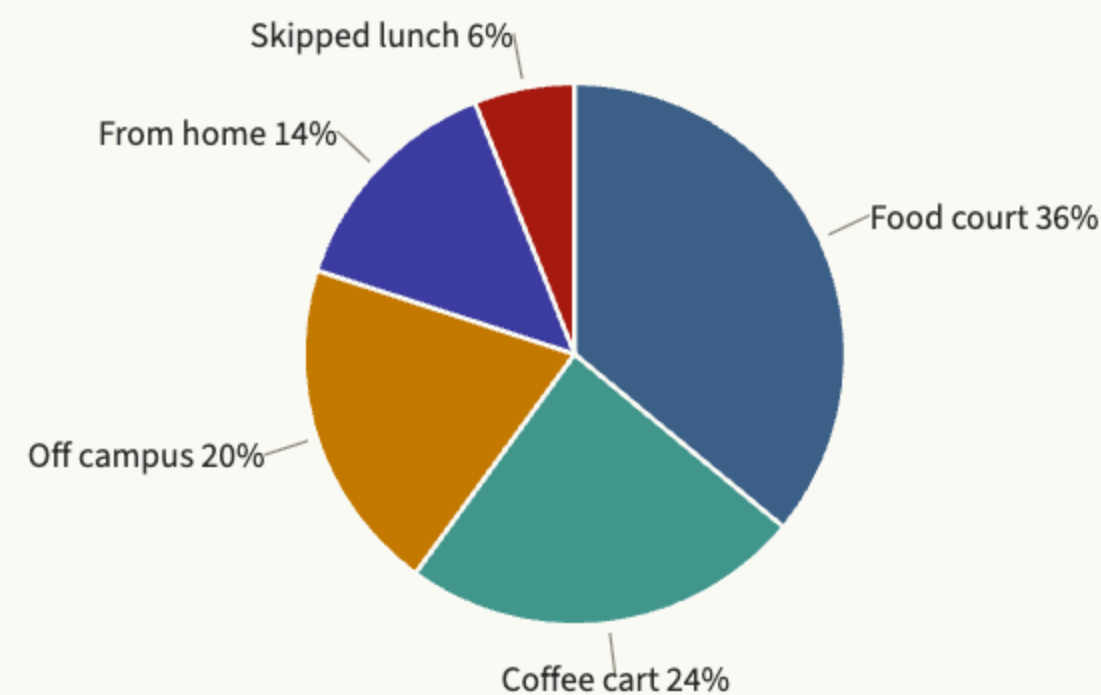
Food court, 36%.

(b) How many ate food from home?

$0.14 \times 250 = 35$ students.

(c) Percent not food court or off campus?

$100 - 36 - 20 = 44\%$.



Pictures That Count

Definition 2.3.28

A **pictograph** is a type of data visualization that uses symbols or pictures to represent data. In a pictograph, each symbol or picture typically represents a certain quantity or value of the data being presented.

Always find the key first. One symbol is rarely one thing.

Library Books by Club

The pictograph shows how many library books each of five student clubs checked out this month.

(a) How many did Robotics check out?

$5\frac{1}{2}$ symbols $\times 4 = 22$ books.

(b) Which club checked out the fewest?

Debate: $1\frac{1}{2}$ symbols, so 6 books.

(c) How many more books did Art check out than Hiking?

$16 - 8 = 8$ more books.

Art 

Chess 

Debate 

Hiking 

Robotics 

 = 4 books

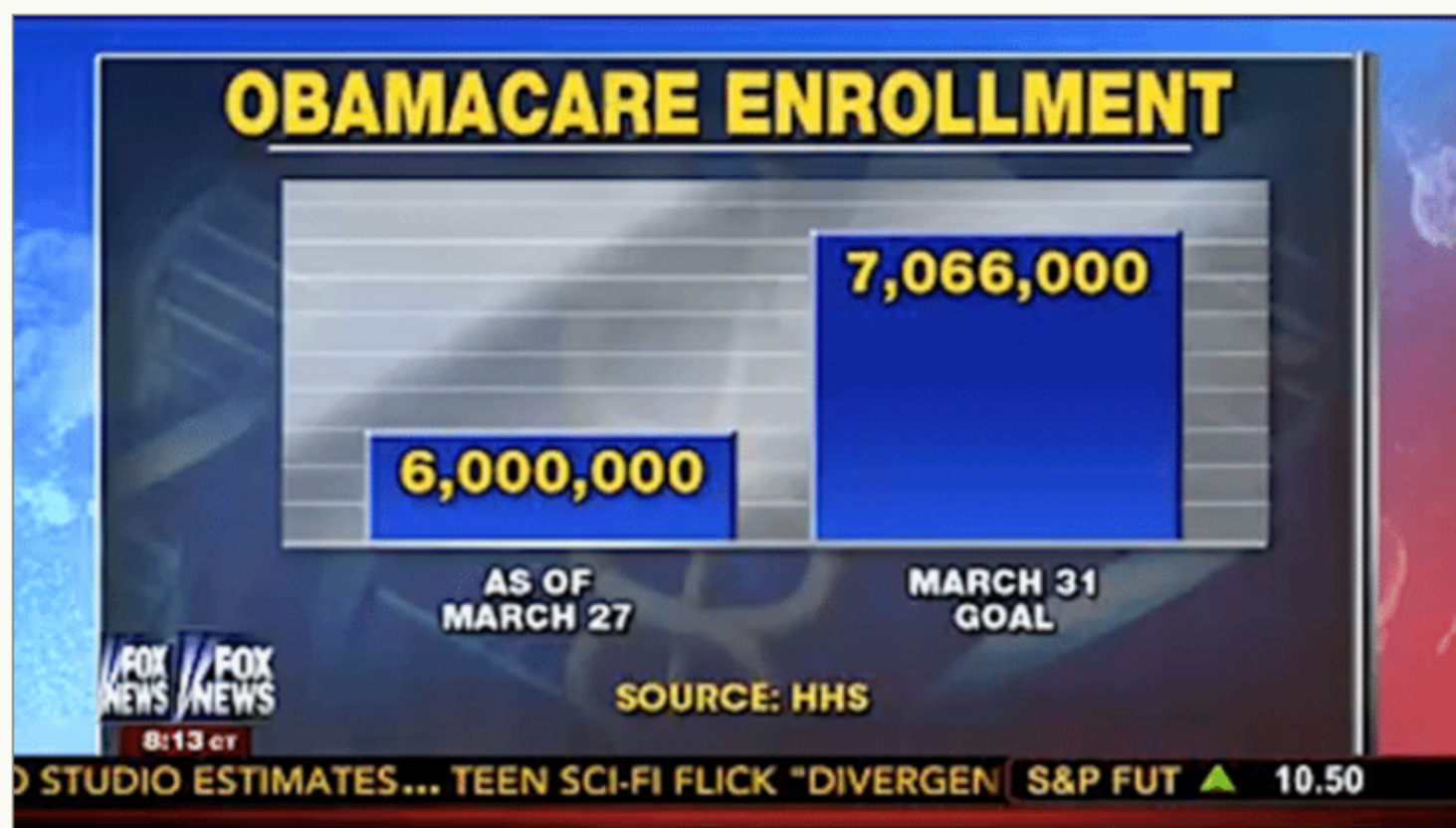
When the Axis Lies

Definition 2.3.38

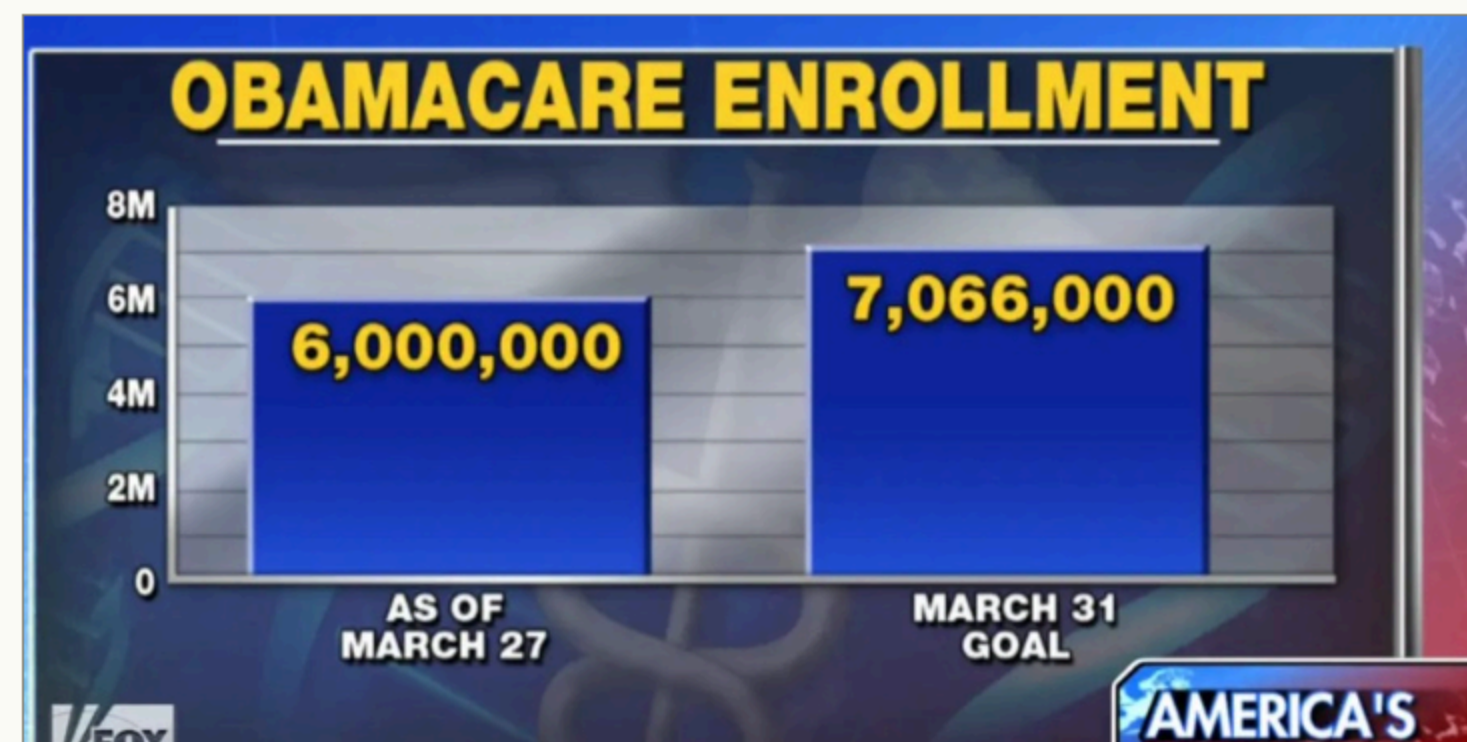
A graph has a **nonzero axis** when one or both of its axes do not begin at 0. Sometimes this is appropriate (for example, to avoid large blank regions in the graph), but a nonzero axis can also be used to misrepresent data by visually exaggerating differences between values.

Not always a trick. The question is whether the missing zero changes what you see.

6.0 Million Against 7.1 Million



As aired on Fox News, 2014. Where does the axis start?



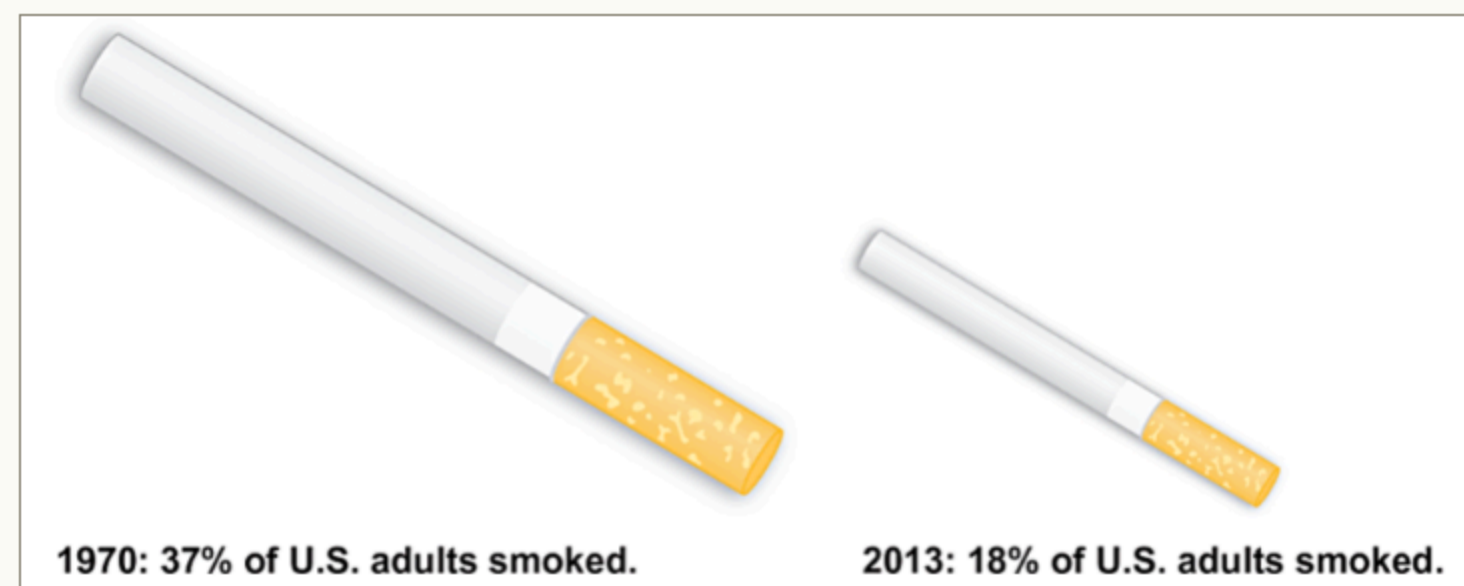
Redrawn from 0. The bars are nearly the same height.

Pictures That Grow Two Ways

Adult smoking fell from 37% in 1970 to 18% in 2013. That is about half.

Does the small cigarette look like half of the big one?

No. It looks like a quarter. The small picture is half as long **and** half as wide. Half of a half is a quarter of the area.

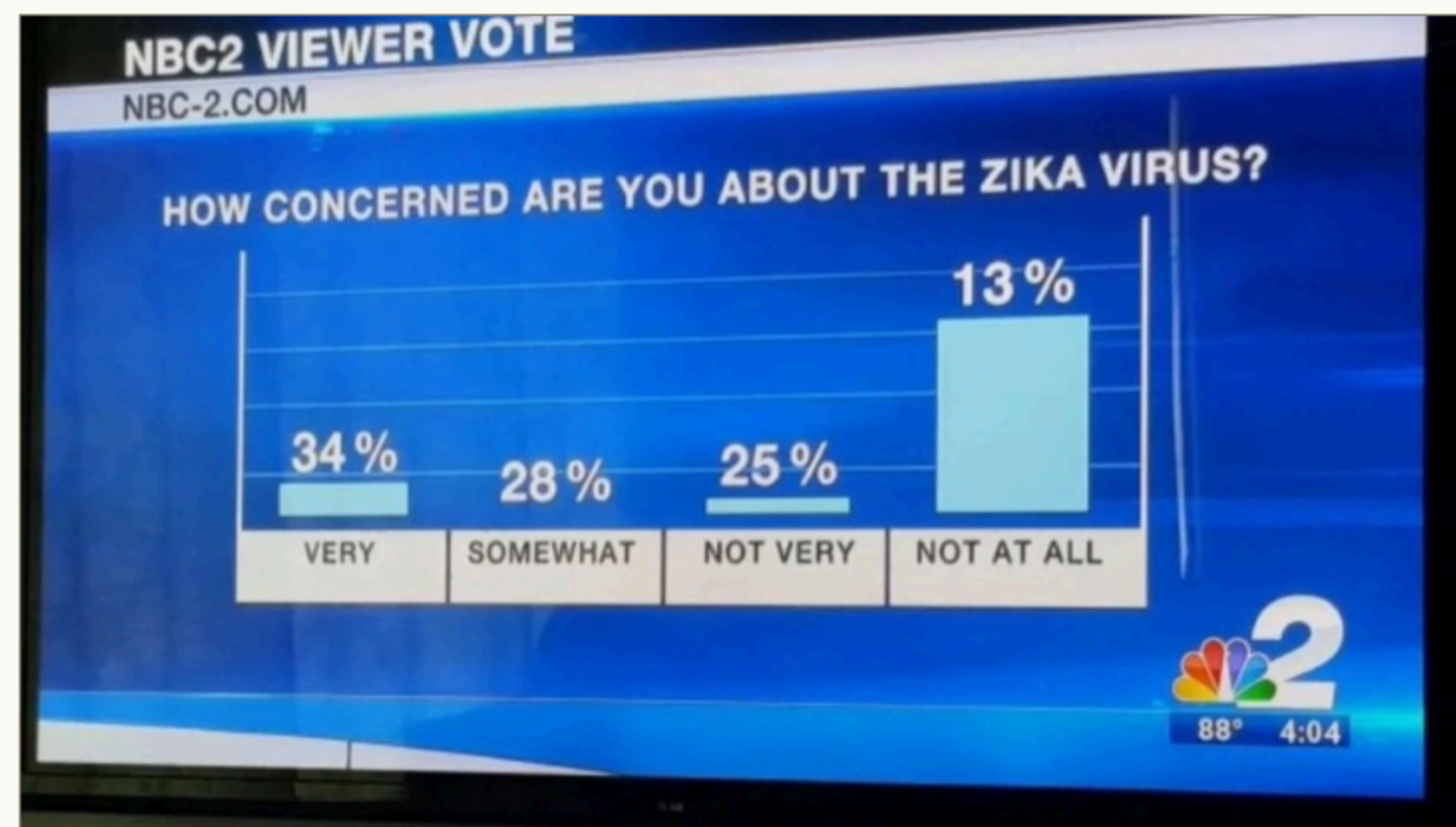


Bars That Ignore Their Own Numbers

A TV station asked viewers how concerned they were about the Zika virus.

What is wrong with this graph?

The bars do not match the numbers. 13% gets the tallest bar and 34% the shortest. Read the numbers, not the bars.



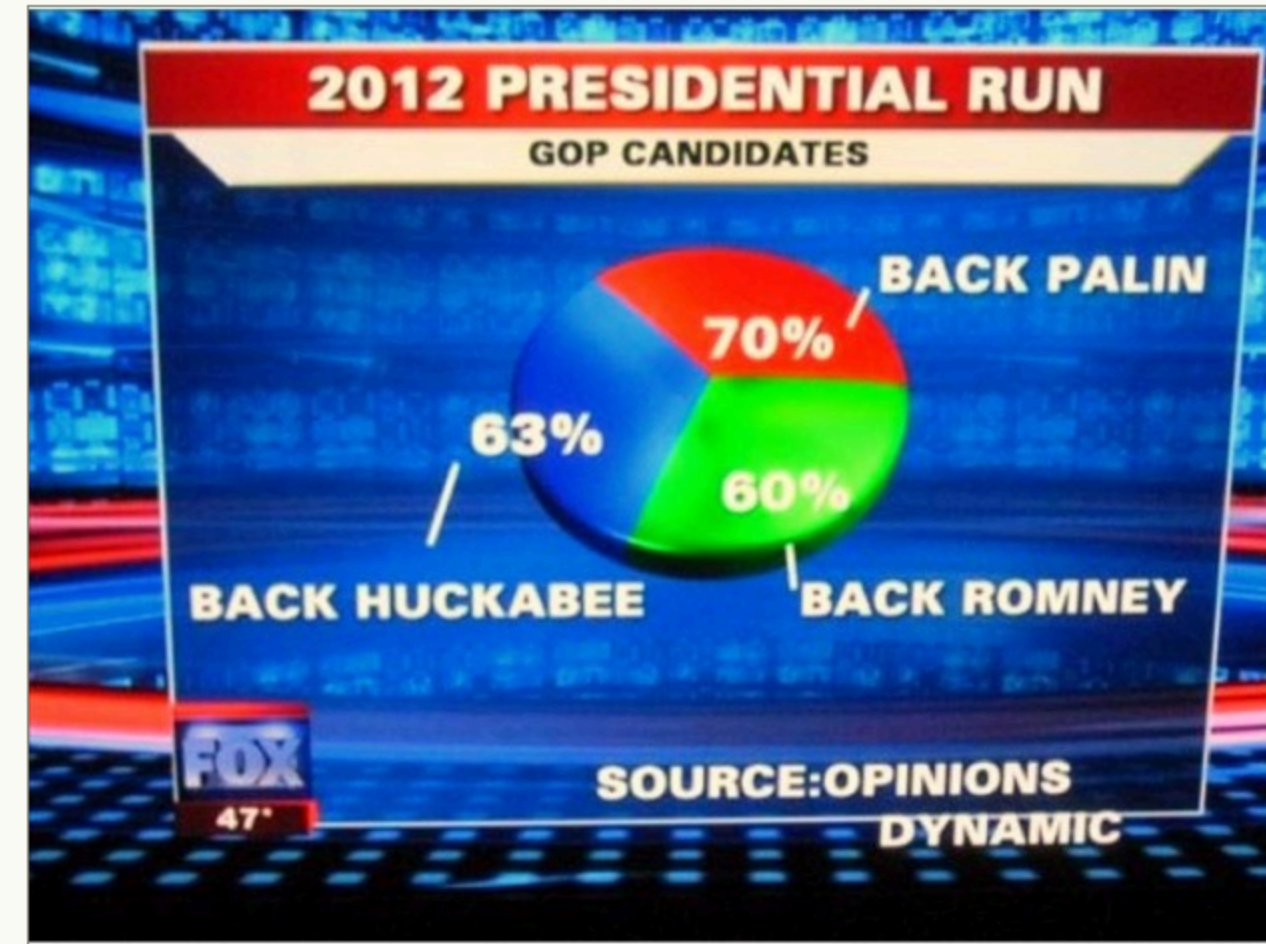
As aired on NBC2, a Fort Myers station.

A Pie That Adds Up to 193%

A TV poll of Republican voters before the 2012 presidential primaries.

What is wrong with this pie chart?

$70\% + 63\% + 60\% = 193\%$. A pie is one whole, so its slices must add to 100%. Voters could back more than one candidate, so this is not pie chart data at all.



As aired on Fox News.

Which Categories, and Since When?

A faculty union graph of how CSU spending has changed since 2019.

The scale is honest. What is the problem?

What counts as “Management Costs”? The categories are never defined, so nobody can check what is in each bar.

And why 2019? The starting year is a choice. A different start year can tell a different story.



From the California Faculty Association.

Five Tricks to Watch For

1. **Nonzero axis.** Starting the vertical axis above zero exaggerates differences.
2. **Inconsistent scales.** Two graphs meant to be compared, drawn on different scales.
3. **3D effects.** Depth and perspective make heights and slices hard to compare.
4. **Pictograph scaling.** Growing a picture in both height and width distorts the ratio.
5. **Omitting data.** Showing only the years, or the categories, that make the point.

Sometimes it is an accident. Sometimes it is not. Either way, your job is to notice.

Two Numbers From Each Person

Definition 2.3.33

Paired data (also called **bivariate data**) consists of two quantitative values collected from the same individual or object. Each pair of values (x, y) represents:

- The **explanatory variable** (or **independent variable**) x : the variable that may explain or predict the other
- The **response variable** (or **dependent variable**) y : the variable that may be affected or predicted

One Point Per Person

Definition 2.3.34

A **scatter plot** (or **scatterplot**) is a graph that displays paired data as points on a coordinate plane. Each point represents one individual:

- The x -coordinate corresponds to the explanatory variable
- The y -coordinate corresponds to the response variable

Scatter plots reveal patterns, trends, and the form of the relationship between two variables.

Practice Minutes and Free Throws

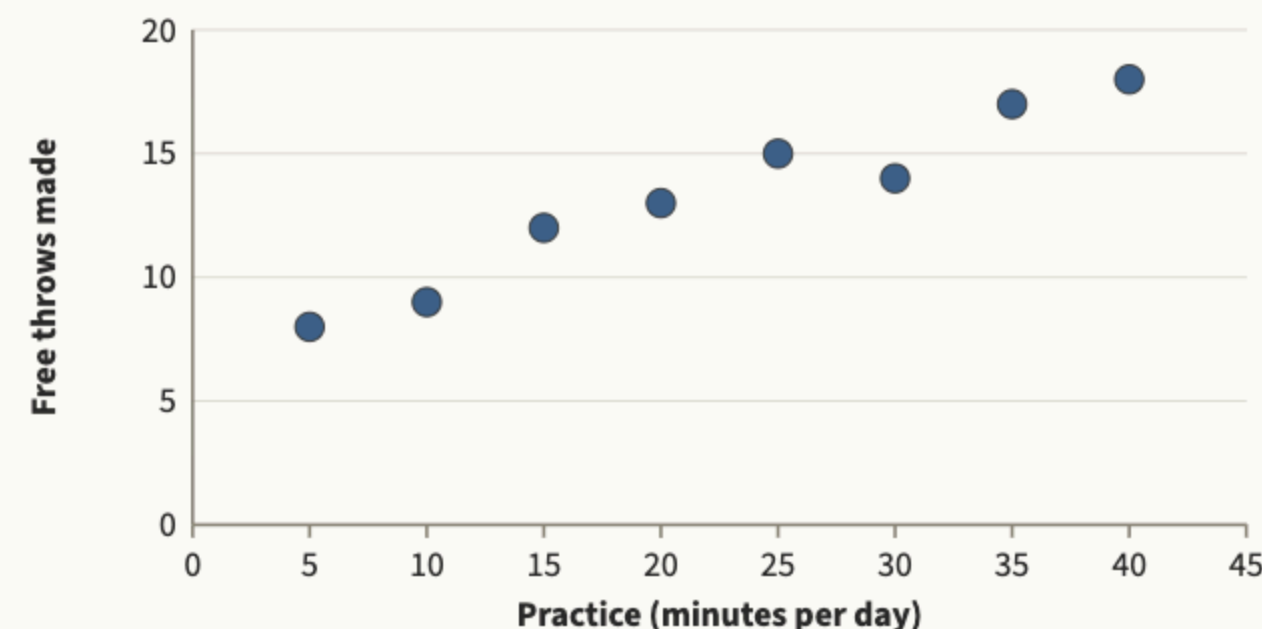
Practice (minutes a day)	5	10	15	20	25	30	35	40
Free throws made (of 20)	8	9	12	13	15	14	17	18

(a) Which variable is explanatory?

Practice. It may explain the score, so it is x .

(b) What pattern do the points show?

They rise together. Players who practice more tend to make more free throws. Chapter 12 measures how strongly.



Now You Try: Pick the Picture

You Try 2.3.3. Match each data set to its best display.

Stem-and-leaf, histogram, time-series, bar graph, pie chart, scatter plot. One is never used.

The data, and what you want to show	Display
Ages of 15 book club members, every age kept	Stem-and-leaf plot
A school budget split among departments	Pie chart
Park visitors, counted monthly for three years	Time-series graph
Birth weights of 500 newborns, in classes	Histogram
Favorite pizza topping of 80 students	Bar graph

Let Excel Draw It

Excel makes today's charts from a two-column table. The click-by-click steps are in the notes:

- Procedure 2.3.8, Creating a Line Graph in Excel
- Procedure 2.3.18, Creating a Bar Graph in Excel
- Procedure 2.3.25, Creating a Pie Chart in Excel

Every one starts the same way: two columns, select both with the headers, Insert tab, pick the chart.